

1. Summarising Notes

Prompt(Using prompt chaining) :

Step-1: Read the Python notes that is attached and list the major topics covered

AI Response:

Main topics covered

- String Slicing
- String Split Method
- Classes and Objects
- Variables in Classes
- Inheritance
- Polymorphism
- Operator Overloading
- Dynamic Attributes
- Constructors and Destructors
- Exception Handling
- Built-in Exceptions
- Custom Exceptions
- File Handling
- File Modes
- File Paths
- seek() Method
- with Statement
- Tkinter GUI
- Tkinter Widgets
- Event Handling
- Mouse Events
- Frames and Menus
- Unit Testing
- Test Cases
- Test Suites
- Assertions
- unittest Module

Step-2: Based on these topics, list the important points under each topic.

AI Response:

Important Points Under Each Topic

- **Strings:** Slicing, split(), reverse strings.
- **OOP:** Classes, objects, variables, constructors, destructors.
- **Inheritance:** Single, Multilevel, Hierarchical, Multiple, Hybrid.

- **Polymorphism:** Overriding, Overloading, Operator Overloading.
- **Dynamic Attributes:** setattr(), getattr().
- **Exception Handling:** try, except, else, finally, raise.
- **Built-in Exceptions:** ZeroDivisionError, ValueError, TypeError, etc.
- **Custom Exceptions:** User-defined exceptions.
- **File Handling:** Open, Read, Write, Close files.
- **File Modes:** r, w, a, r+, w+, x, b, t.
- **File Utilities:** seek(), paths, with statement.
- **Tkinter:** Windows, widgets, events.
- **Widgets:** Label, Button, Entry, Text, Radiobutton, Checkbutton.
- **Event Handling:** command, bind(), mouse events.
- **Unit Testing:** Test cases, assertions, unittest module.

Step-3: Using the important points identified above, write a short summary for a B.Tech student.

AI Response:

Python Mid-2 Notes – Short Summary for B.Tech Students

Python Unit-3 focuses on **Object-Oriented Programming (OOP)** and **String Manipulation**. Students learn how to work with strings using slicing and split methods. The unit introduces OOP concepts such as **classes, objects, class variables, instance variables, constructors, destructors, dynamic attributes, inheritance, and polymorphism**. Different types of inheritance (single, multilevel, hierarchical, multiple, and hybrid) are discussed along with method overriding, method overloading, and operator overloading.

Python Unit-4 covers **Exception Handling** and **File Handling**. Students learn about different types of errors, built-in exceptions, custom exceptions, and the use of **try, except, else, finally, and raise** statements for handling runtime errors. The file handling section explains how to open, read, write, append, and close files using different file modes. It also covers text and binary files, file paths, the seek() method, and the **with statement** for efficient resource management.

Python Unit-5 introduces **Graphical User Interface (GUI) development using Tkinter** and **Unit Testing**. Students learn to create windows and use widgets such as Labels, Buttons, Entry boxes, Text boxes, Checkbuttons, and Radiobuttons. The unit also explains event-driven programming, event handling, mouse events, frames, and menus. Finally, it covers software testing concepts including test cases, test suites, assertions, and the **unittest module** for writing and running automated tests.

Overall, these units help B.Tech students understand **OOP principles, error handling, file operations, GUI application development, and software testing**, which are essential skills for Python programming and software development.

Why this worked:

The task was divided into multiple connected prompts. Each response was used as input for the next prompt, helping the AI generate a more organized and accurate summary.

2. Generating a Study Plan

Prompt:

Act as a Python teacher. Create a 7-day study plan for learning Python basics. For example, include one topic for each day. Use simple language and give the output in a day-wise format.

AI Response:

7-Day Python Basics Study Plan

Day 1: Introduction to Python

- What is Python?
- Installing Python
- Writing your first Python program
- Using `print()` function

Day 2: Variables and Data Types

- Variables
- Numbers, Strings, and Booleans
- Taking user input with `input()`

Day 3: Operators

- Arithmetic Operators
- Comparison Operators
- Logical Operators
- Simple programs using operators

Day 4: Decision Making

- `if`, `elif`, and `else`
- Nested `if` statements
- Practice examples

Day 5: Loops

- `for` loop

- while loop
- break and continue
- Pattern programs

Day 6: Functions

- Defining functions using def
- Function arguments
- Return statement
- Simple function programs

Day 7: Lists and Strings

- List creation and operations
- String operations
- String slicing
- Small practice programs

Goal: By the end of 7 days, you will understand Python fundamentals and be ready to learn advanced topics like OOP, File Handling, and Exception Handling.

Why this worked:

This prompt worked because it gave clear instructions about what to create, how to create it, and the format of the output.

3. Creating Interview Questions

Prompt:

Act as a technical interviewer. Generate 10 beginner level Python interview questions. For example, include questions on variables, loops, and functions. Use a simple and professional tone. Present the questions in a numbered list.

AI Response:

1. What is Python, and why is it popular among programmers?
2. What is a variable in Python? Give an example.
3. What are the basic data types available in Python?
4. What is the difference between a string and an integer?
5. How do you take input from a user in Python?
6. What is the purpose of an if statement? Write a simple example.
7. What is the difference between a for loop and a while loop?
8. How do you create and use a function in Python?
9. What is a list in Python? How do you access its elements?
10. What is the difference between break and continue statements in loops?

Why this worked:

This prompt worked because it clearly specified the role, topic, number of questions, and output format, helping the AI generate relevant interview questions.

4. Writing a Resume Summary

Prompt:

Act as a professional resume writer. Write a short resume summary for a B.Tech CSE student. For example, include skills like Python, MERN Stack, and Machine Learning. Use a professional tone and present the summary in one paragraph.

AI Response:

Motivated B.Tech Computer Science and Engineering student with a strong foundation in Python programming, MERN Stack development, and Machine Learning. Skilled in problem-solving, web application development, and building data-driven solutions. Eager to apply technical knowledge to real-world projects, learn emerging technologies, and contribute effectively to innovative software development teams.

Why this worked:

This prompt worked because it clearly mentioned the student's skills, the role, and the required format, helping the AI generate a professional resume summary.

5. Creating a Lesson Plan

Prompt:

Act as a professional resume writer. Write a short resume summary for a B.Tech CSE student. For example, include skills like Python, MERN Stack, and Machine Learning. Use a professional tone and present the summary in one paragraph.

AI Response:

One-Hour Python Basics Lesson Plan

Time	Topic
0 – 10 min	Introduction to Python- What is Python?- Features and applications of Python- Writing the first Python program using print()

Time	Topic
10 – 20 min	Variables and Data Types- Variables- Integer, Float, String, Boolean- Taking user input using input()
20 – 30 min	Operators- Arithmetic Operators (+, -, *, /)- Comparison Operators (==, >, <)- Logical Operators (and, or, not)
30 – 40 min	Decision Making- if, elif, else statements- Simple examples and practice problems
40 – 50 min	Loops- for loop- while loop- break and continue statements
50 – 60 min	Functions- Creating functions using def- Function parameters and return values- Simple function examples

Learning Outcome

By the end of this one-hour session, students will be able to write basic Python programs using variables, operators, conditions, loops, and functions.

Why this worked:

This prompt worked because it clearly mentioned the topic, role, duration, and format, helping the AI generate a well-structured lesson plan.